

Optimized algorithm: geodesic edge-labeling of the ring C_{10}

Ring C_{10} with 10 edges · geodesic weights = $\{1, \dots, 50\}$

1 Formalization

Cycle C_{10} with edges $e_0, \dots, e_9$ (indices taken mod 10) and weights

$$w_i \in \mathbb{Z}^+, \quad i \in \mathbb{Z}_{10} = \{0, \dots, 9\}.$$

For a start s and length $L \in \{1, \dots, 5\}$ the window sum is

$$W(s, L) = \sum_{k=0}^{L-1} w_{(s+k) \bmod 10}.$$

Geodesics of C_{10} are exactly the windows of length 1 to 5 (length-5 diametric pairs contribute two shortest arcs), so the geodesic set is

$$\mathcal{G} = \{ W(s, L) : s \in \mathbb{Z}_{10}, L \in \{1, \dots, 5\} \}, \quad |\mathcal{G}| = 10 \times 5 = 50.$$

Goal: find $w = (w_0, \dots, w_9)$ with $\mathcal{G} = \{1, 2, \dots, 50\}$
(a bijection, since $|\mathcal{G}| = 50$).

2 Necessary conditions (pruning invariants)

Each edge occurs in $1+2+3+4+5 = 15$ windows, hence

$$\sum_{s, L} W(s, L) = 15 \sum_i w_i = \sum_{n=1}^{50} n = 1275 \Rightarrow S := \sum_i w_i = 85. \quad (\text{C1})$$

$1 \in \mathcal{G}$ forces some $w_i = 1$ (any $L \geq 2$ sum is $\geq 1+2 = 3$);

$2 \in \mathcal{G}$ forces some $w_j = 2$ (since $2 = 1+1$ is barred by distinctness).

$$\exists i, j: w_i = 1, w_j = 2. \quad (\text{C2})$$

Opposite length-5 arcs partition the edge set, so $W(s, 5) + W(s+5, 5) = S = 85$;
with both ≤ 50 this gives

$$35 \leq W(s, 5) \leq 50 \quad \forall s. \quad (\text{C3})$$

Distinct windows force distinct w_i ; combined with (C1),

$$w_i \leq S - \sum_{k=1}^9 k = 85 - 45 = 40 \quad \forall i. \quad (\text{C4})$$

Rotational symmetry is quotiented by pinning the unit edge:

$$w_0 : = 1.$$

(C5)

3 The algorithm (pruned depth-first search)

State: a partial assignment $w_0, \dots, w_{i-1}$, running sum

$\sigma = \sum_{k < i} w_k$, and the set $U \subseteq \{1, \dots, 50\}$ of geodesic

values already realized by non-wrapping windows $W(s, L)$ with $s + L \leq i$.

Placing value v at position i fails iff some newly-completed window $W(i-L+1, L)$ violates one of the necessary clauses:

$$(W > 50) \vee (W \in U) \vee (L = 5 \wedge W < 35) \vee (\sigma + v > 85).$$

Every clause is necessary, so pruning is sound (no valid labeling is discarded) and the enumeration is complete.

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Search(i, sigma, U):
  if i = 10:
    report solution if {W(s,L) : s in Z10, L in 1..5} = {1..50}
    return # includes wrapping windows
  for v = 1 ... 40 with v not in {w0..w_{i-1}} and sigma+v <= 85:
    w_i <- v ; A <- {} ; ok <- true
    for L = 1 ... min(5, i+1): # windows ending at i
      W <- sum_{t=i-L+1}^{i} w_t
      if W>50 or W in U or (L=5 and W<35): ok<-false; break
      A <- A + {W}
    if ok: Search(i+1, sigma+v, U + A)
    undo w_i, A

Driver: w0 <- 1 (C5); Search(1, 1, {1}).
```

An optional reflection quotient ($w_1 < w_9$, checked once w_9 is placed) halves the tree.

4 Complexity

Naive enumeration of 10 distinct values in $[1, 40]$ around the ring is

$$\binom{40}{10} \cdot \frac{9!}{2} \approx 10^{16} \quad (\text{infeasible}).$$

The predicate collapses this. With (C1) to (C5) the entire tree is

384 nodes, depth < 10 on every branch,

i.e. a few hundred visited states, decidedly not brute force. Dropping (C3) and the tight cap still finishes in 7.4×10^5 nodes; dropping all symmetry and bounds, 1.4×10^7 .

5 Result

$\mathcal{G} = \{1, \dots, 50\}$ is unsatisfiable on C_{10} .

Confirmed by four independent methods:

method	assumptions	nodes	sols
pruned DFS	WLOG symm. + bound	735,496	0
unconstrained DFS	none	14,373,810	0
necessary-cond. tree	(C1)-(C5)	384	0
Z3 SMT	direct model	unsat	0

Under (C1)-(C5) no branch reaches depth 10: every path ends in a range violation ($W > 50$), a value collision ($W \in U$), or a (C3) violation ($W < 35$).